Jie Shen

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Education

University of Toronto *September 2022 – April 2027*
Honours B.Sc. in Mathematics and Statistics, Minor in Computer Science Mississauga, ON
Relevant Coursework: Machine Learning, Statistical Learning, Regression Analysis, Data Structures & Algorithms, Software Design, Probability, Statistical Inference

Technical Skills

Languages: Python, Java, JavaScript, R, SQL, HTML, CSS
Frameworks & Tools: FastAPI, FAISS, Scikit-Learn, NumPy, Pandas, JavaFX, JUnit, Git, GitHub, LaTeX
Core Skills: Object-Oriented Programming, MVC, REST APIs, NLP, Classification, Regression, Cross-Validation, Feature Engineering, Data Structures & Algorithms

Projects

AI Study Assistant *Summer 2026*
Python, FastAPI, FAISS, SentenceTransformers, PyMuPDF, Ollama
- Built a **Retrieval-Augmented Generation (RAG)** application that processes uploaded PDFs and answers natural-language questions using retrieved document context.
- Implemented a modular document pipeline using **PyMuPDF extraction**, overlapping text chunking, SentenceTransformers embeddings, cosine-similarity retrieval, and FAISS vector search.
- Designed a FastAPI backend with document-upload and question-answering endpoints, configurable retrieval parameters, and clear separation between document processing, retrieval, and response generation.
- Created a provider abstraction supporting **local Ollama/Llama 3.2 inference** and optional cloud-based LLM providers without changing the surrounding retrieval pipeline.

Interactive Canvas Studio *Summer 2025; Refactored in 2026*
Java, JavaFX, MVC-Style Architecture, Factory Pattern, Object-Oriented Design
- Developed a JavaFX desktop drawing application with **8 drawing tools**, real-time mouse-driven rendering, colour and fill controls, adjustable line thickness, and light/dark display modes.
- Designed an extensible **MVC-style architecture** that separates application state, rendering, and user-input handling into modular model, view, and controller components.
- Implemented a polymorphic shape hierarchy with factory-based creation for circles, rectangles, squares, ovals, triangles, polylines, freehand sketches, and wave tools.
- Built undo/redo, cut/copy/paste, and custom `.canvas` save/load persistence while maintaining modular components and architecture documentation for future feature expansion.

Hybrid NLP & Behavioural Classification Engine *Winter 2026*
Python, NumPy, Pandas, NLP, Multinomial Logistic Regression, Machine Learning
- Built an end-to-end **3-class classification system** combining Bag-of-Words text representations with normalized behavioural survey variables.
- Implemented multinomial logistic regression **from scratch** using vectorized gradient descent, softmax activation, cross-entropy loss, and L2 regularization.
- Achieved approximately **88% multiclass classification accuracy** through hybrid feature engineering, hyperparameter tuning, and comparison with alternative classification models.
- Structured the system into reusable preprocessing, feature engineering, training, inference, tuning, and evaluation modules to support reproducible experimentation.

Experience

City of Mississauga *July 2018 – March 2019*
Summer Camp Leader Mississauga, ON
- Coordinated daily activities and supervised participant groups while collaborating with staff in a fast-paced public recreation environment.
- Strengthened communication, leadership, and organization through activity planning, participant engagement, and issue resolution.